🔷 A.11 — Primal–Dual Convergence (Final — Rigorous Version)

A.11.1 Constrained Problem

$$\min_{\Psi \in \Omega}\;\mathcal{E}(\Psi)
\quad \text{s.t.} \quad g(\Psi)=0$$

Assumptions:
- $\mathcal{E}$ is convex and differentiable
- $g(\Psi)=A\Psi-b$ is affine (linear constraints)
- Ω is compact and convex

A.11.2 Lagrangian

$$\mathcal{L}(\Psi,\lambda)
=
\mathcal{E}(\Psi) + \lambda^T g(\Psi)$$

A.11.3 Projected Primal–Dual Dynamics

$$\dot{\Psi}
=$$

$$\Pi_{\Omega}\big(\Psi,\; -\nabla_{\Psi}\mathcal{L}(\Psi,\lambda)\big)$$

$$\dot{\lambda} = g(\Psi)$$

where $\Pi_{\Omega}(x,v)$ is the projection of direction v onto the tangent cone of Ω at x.

A.11.4 KKT Equilibrium

At equilibrium (Ψ^*,λ^*):

$$\nabla \mathcal{E}(\Psi^*) + A^T \lambda^* = 0, \quad g(\Psi^*) = 0$$

Thus:

KKT conditions hold.

A.11.5 Lyapunov Function

$$V(\Psi,\lambda)$$

$$
= \mathcal{E}(\Psi) - \mathcal{E}(\Psi^*) + \frac{1}{2}\|\lambda - \lambda^*\|^2
$$

Since $\mathcal{E}$ is convex:

$$
V(\Psi,\lambda) \ge 0
$$

A.11.6 Time Derivative

$$
\dot{V} = \langle \nabla \mathcal{E}(\Psi), \dot{\Psi} \rangle + (\lambda - \lambda^*)^T \dot{\lambda}
$$

Substitute dynamics:

$$
\dot{V} = \langle \nabla \mathcal{E}, -\nabla_{\Psi} \mathcal{L} \rangle + (\lambda - \lambda^*)^T g(\Psi)
$$

A.11.7 Key Inequality (Convexity + Saddle

Property)

Using:

$$\nabla_{\Psi}\mathcal{L} = \nabla \mathcal{E} + A^T \lambda$$

we obtain:

$$\dot{V} = - \|\nabla_{\Psi}\mathcal{L}\|^2 + (\lambda - \lambda^*)^T g(\Psi)$$

From convexity of $\mathcal{E}$ and linearity of g, the Lagrangian is convex in Ψ and concave in λ, hence:

$$(\lambda - \lambda^*)^T g(\Psi) \le 0$$

Therefore:

$$\dot{V} \le -\|\nabla_{\Psi}\mathcal{L}\|^2 \le 0$$

A.11.8 Convergence (LaSalle Argument)

Since:

$$\dot{V} \le 0$$

all trajectories remain bounded and converge to

the largest invariant set where:
$$\nabla_{\Psi}\mathcal{L} = 0, \quad g(\Psi)=0$$

Thus:
$$(\Psi,\lambda) \to (\Psi^*,\lambda^*)$$

A.11.9 Constraint Satisfaction

From:
$$\dot{\lambda} = g(\Psi)$$

and convergence:
$$g(\Psi^*) = 0$$

Thus:
$$\Psi^* \in \mathcal{X}_{\mathrm{feasible}}$$

A.11.10 Projection (Safety Layer)

Projection operator:
$$\Psi \leftarrow \Pi_{\Omega}(\Psi)$$

satisfies:
$$\|\Pi_{\Omega}(x) - \Pi_{\Omega}(y)\| \le \|x-y\|$$

Thus:

projection is non-expansive and preserves stability.

A.11.11 Compatibility with SEKHEM-TSDD

- Included in dynamics via -\nabla \mathcal{L}
- Preserves Lyapunov structure
- Compatible with bounded control

FINAL RESULT — A.11

\boxed{
\text{The system converges to a feasible, optimal, constraint-satisfying solution.}
}

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219